t3-w06 — Construction — Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — construct perpendicular lines

`objective_text`

```
construct perpendicular lines
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — bisect a given line segment

`objective_text`

```
bisect a given line segment
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

```
construct angle $90^\circ$
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 5 — construct angle 60°

`objective_text`

```
construct angle $60^\circ$
```

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Constructing Parallel Lines with a Ruler and Set-Square

> EXPLANATION
> Parallel lines are lines in the same plane that never meet, no matter how far they are extended. To draw a line parallel to a given line using a ruler and a set-square, follow these steps:
>
> 1. Place the set-square so that one of its right-angle sides lies along the given line.
> 2. Slide the set-square along the ruler until the other side passes through the point where you want the new line.
> 3. Draw a line along the edge of the set-square that is opposite the given line. This new line will be parallel to the original.

> WORKED_EXAMPLE
> Example: Construct a line through point C that is parallel to line AB.
>
> 1. Draw line AB and mark point C somewhere away from it.
> 2. Place the set-square so that its bottom edge lies on AB.
> 3. Keeping the set-square in place, slide the ruler until the top edge of the set-square passes through C.
> 4. Draw a line along the top edge of the set-square. The line through C is parallel to AB.

1. Given line segment DE and a point F not on the line, construct a line through F that is parallel to DE. Describe the steps you would take.

answer: 1. Place the set-square with one side on DE. 2. Slide the ruler until the opposite side of the set-square passes through F. 3. Draw a line along that opposite side; it will be parallel to DE.

2. You have line GH drawn on your paper and a point I located elsewhere. How would you construct a line through I that is parallel to GH?

answer: 1. Align the set-square so its bottom edge lies on GH. 2. Move the ruler while keeping the set-square fixed until the top edge goes through I. 3. Draw the line along the top edge; this line is parallel to GH.

3. Line JK is given, and point L is off the line. Explain how to draw a line through L that is parallel to JK.

answer: 1. Position the set-square with one side on JK. 2. Slide the ruler until the opposite side of the set-square reaches L. 3. Draw a line along that opposite side; it will be parallel to JK.

Constructing Perpendicular Lines

EXPLANATION

Two lines are perpendicular when they meet to form a right angle (90°). To draw a line that is perpendicular to a given line using only a compass and a straightedge, follow these steps:

1. Draw the given line (or have it already drawn) and mark the point through which the perpendicular line must pass.
2. Place the compass point on that point and draw an arc that cuts the given line at two points.
3. Keeping the same compass width, place the compass point on each of the two intersection points and draw two arcs that intersect each other.
4. Connect the original point with the intersection of the two arcs. The line you draw is perpendicular to the original line.

WORKED_EXAMPLE

Example: Construct a line through point C that is perpendicular to the line AB.

1. With the compass set to a convenient radius, place the needle on C and draw an arc that cuts AB at points D and E.
2. Without changing the radius, place the needle on D and draw an arc below AB.
3. With the needle on E, draw another arc that meets the previous one at point F.
4. Use the straightedge to draw the line CF. Line CF is perpendicular to AB.

1. You have a line segment AB and a point C not on the line. List the steps to construct a line through C that is perpendicular to AB using a compass.

answer: 1. Place the compass point on C and draw an arc intersecting AB at two points D and E. 2. Without changing the radius, place the compass point on D and draw an arc below AB. 3. With the compass point on E, draw another arc intersecting the previous arc at point F. 4. Draw a straight line through C and F; this line is perpendicular to AB.

2. When constructing a perpendicular line to a given line using a set square, which side of the set square should be placed against the given line?

answer: The side that forms a right angle (the 90° corner) should be placed against the given line.

3. If you need a perpendicular line through point P on line XY and you have a protractor, what angle should you mark at point P?

answer: Mark a 90° angle at point P.

How to Bisect a Line Segment

> EXPLANATION
> To bisect a line segment you need only a ruler and a compass.
>
> 1. **Set the compass** to a radius larger than half the length of the segment.
> 2. **Place the compass point on one endpoint** of the segment and draw an arc above and below the line. This creates two points where the arc meets the paper.
> 3. **Without changing the radius**, move the compass to the other endpoint and draw another arc that intersects the first two arcs. You will now have two intersection points – one above and one below the original segment.
> 4. **Draw a straight line** through the two intersection points. This line cuts the original segment exactly at its midpoint, thereby bisecting it.

> WORKED_EXAMPLE
> **Worked Example – Bisecting segment AB**
>
> - Choose a radius larger than half of AB. With the compass point at **A**, draw arcs above and below AB, creating points **P** (above) and **Q** (below).
> - Keeping the same radius, place the compass point at **B** and draw arcs that intersect the first arcs. These intersect at points **R** (above) and **S** (below).
> - The two intersection points above the segment are **P** and **R**; the two below are **Q** and **S**.
> - Draw the line **PR** (or **QS**). The point where this line meets AB is the midpoint **M**. Segment AB is now bisected at M.

1. Given a line segment CD, describe how to bisect it using a compass and straightedge.
answer: Draw arcs from C and D with the same radius larger than half CD, intersecting above and below the segment. Connect the two intersection points; the line meets CD at its midpoint, bisecting it.

2. You have a segment EF that is 8 cm long. Using a compass, construct its midpoint. List the construction steps.
answer: Set the compass to a radius greater than 4 cm. With the point at E, draw arcs above and below EF. Without changing the radius, move to F and draw arcs intersecting the first ones. Connect the two intersection points; the crossing point on EF is the midpoint.

3. Bisect the line segment GH and label the midpoint M.
answer: Draw arcs from G and H with equal radius, intersecting at two points. Draw the line through these points; it meets GH at M, the midpoint.

Constructing a Right Angle (90°)

EXPLANATION

A right angle measures exactly 90°, meaning the two lines are perpendicular. You can create a right angle at a point on a line using only a ruler and a compass. The method works by copying a 90° angle from a known right angle (the corner of a square or the intersection of two perpendicular lines) onto the given line.

Steps to construct a 90° angle at point P on line l

1. Place the compass point at P and draw an arc that cuts line l at two points, A and B.
2. Without changing the compass width, place the compass point at A and draw an arc above the line.
3. Keeping the same width, place the compass point at B and draw another arc that intersects the first arc at point C.
4. Draw a straight line from P through C. This new line is perpendicular to line l, forming a 90° angle at P.

WORKED_EXAMPLE

Worked Example: Construct a right angle at point B on the line AB.

1. Set the compass to a convenient radius and place the needle at B. Draw an arc that meets AB at points D and E.
2. Keeping the same radius, place the needle at D and draw an arc above the line.
3. Move the needle to E and draw another arc that cuts the previous one at point F.
4. Use the ruler to draw a line from B through F. The angle between this new line and AB is 90°.

You can check your construction by measuring the angle with a protractor; it should read 90°.

1. Given a straight line XY, construct a 90° angle at point Y using a ruler and compass. Describe each step you would take.

answer: 1. Place the compass point at Y and draw an arc that cuts XY at two points, M and N. 2. Without changing the compass width, place the compass point at M and draw an arc above the line. 3. Keeping the same width, place the compass point at N and draw another arc that meets the first arc at point O. 4. Draw a straight line from Y through O. The line YO is perpendicular to XY, forming a right angle at Y.

2. You have a line CD. Using only a ruler and compass, construct a right angle at point C. List the construction steps.

answer: 1. Put the compass point on C and draw an arc that cuts CD at points P and Q. 2. Keep the same compass width, place the point on P and draw an arc above the line. 3. With the same width, place the point on Q and draw another arc that intersects the previous one at R. 4. Join C to R with a straight line. The line CR is perpendicular to CD, giving a 90° angle at C.

3. A line segment EF is drawn. Choose a point G that is not on EF and construct a 90° angle at G that meets EF. Explain how you would do this.

answer: 1. From G, draw a line that roughly aims toward EF; let it intersect EF at H. 2. Place the compass point at H and draw an arc that cuts the line GH at two points, I and J. 3. Without changing the radius, place the compass point at I and draw an arc on the opposite side of GH. 4. With the same radius, place the compass point at J and draw another arc that meets the previous one at K. 5. Draw a straight line from G through K. The line GK is perpendicular to GH, so the angle between GK and GH at G is 90°.

Constructing a 60° Angle

EXPLANATION

A 60° angle can be created using only a ruler and a compass by exploiting the properties of an equilateral triangle. All three interior angles of an equilateral triangle are exactly 60°, so if you build such a triangle on a given line, the angle at the base of the triangle will be the required 60° angle.

Key idea: Construct an equilateral triangle on one side of a line; the angle formed between the original line and the side of the triangle is 60°.

Why it works: In an equilateral triangle, each side is equal in length and each interior angle is 60°. When the triangle shares a side with the given line, the angle between that side and the adjacent side of the triangle must be 60°.

WORKED_EXAMPLE

Worked Example: Constructing a 60° angle at point P on line AB

1. **Draw the given line** AB and mark the point P on it where the angle is to be constructed.
2. **Set the compass width** to any convenient length (larger than half the length of AB) and place the compass point on P.
3. **Draw an arc** that cuts the line AB at two points; label the intersection on the left of P as D and on the right as E.
4. **Without changing the compass width**, place the compass point on D and draw an arc above the line.
5. **Again without changing the width**, place the compass point on E and draw another arc that intersects the arc drawn from D. Label the intersection point as F.
6. **Draw a straight line** from P through F. The angle ∠DPF (or ∠EPF) is exactly 60° because PD = PE = PF, forming an equilateral triangle P-D-F (or P-E-F).
7. **Check**: The triangle P-D-F is equilateral, so each interior angle, including the one at P, measures 60°.

You have now constructed a precise 60° angle at point P.

1. Construct a 60° angle at point C on the line CD. List the steps you would follow.

answer: 1. Draw line CD and mark point C. 2. Set the compass to any convenient radius and place the point on C. 3. Draw an arc cutting CD at two points; call them D_1 (left of C) and D_2 (right of C). 4. Without changing the radius, place the compass on D_1 and draw an arc above the line. 5. With the same radius, place the compass on D_2 and draw another arc that meets the first one; label the intersection F. 6. Draw a line from C through F. The angle between CD and CF is 60° because triangle C-D_1-F (or C-D_2-F) is equilateral.

2. You draw an equilateral triangle XYZ on the base XY. What is the measure of each base angle at X and Y?

answer: 60°

3. Using a compass, draw an arc with centre A that meets line AB at points D and E. From D and E draw arcs that intersect at F. What is the measure of angle DAF?

answer: 60°

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Construction

`subtopic`

Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

`term`

Third Term

`week`

6

`lesson_objectives`

• construct parallel lines
• construct perpendicular lines
• bisect a given line segment
• construct angle 90°
• construct angle 60°

`entry_behaviour`

• Students have been taught how to draw lines.

`instructional_materials`

• Mathematical set.

Objective 1 — construct parallel lines

`objective_text`

construct parallel lines

`current_step`

1

`instructional_events`

- present_content
- elicit_performance

`student_tasks`

- construct parallel lines

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — construct perpendicular lines

`objective_text`

construct perpendicular lines

`current_step`

2

`instructional_events`

- present_content

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — bisect a given line segment

`objective_text`

bisect a given line segment

`current_step`

3

`instructional_events`

- present_content
- elicit_performance

`student_tasks`

- construct line segment

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 4 — construct angle 90°

`objective_text`

construct angle 90°

`current_step`

4

`instructional_events`

- present_content
- elicit_performance

`student_tasks`

- construct angle 90 degrees

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 5 — construct angle 60°

`objective_text`

construct angle 60°

`current_step`

5

`instructional_events`

• present_content
• elicit_performance

`student_tasks`

• construct angle 60 degrees

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, source_figures, objective_learner_state.`

Output — the pack it produced (draw 3)

Objective 1

Constructing Parallel Lines

> `EXPLANATION`
> Parallel lines are lines in the same plane that never meet, no matter how far they are extended. To draw a line parallel to a given line using a set square:
>
> 1. Place the set square so that one of its edges lies along the given line.
> 2. Slide the set square until the opposite edge passes through the point where you want the new line.
> 3. Draw a line along this opposite edge. The line you draw will be parallel to the original line.

> `WORKED_EXAMPLE`
> Example: Construct a line through point C that is parallel to line AB.
>
> - Place the set square with one edge on AB.
> - Move the set square until the opposite edge goes through C.
> - Draw a line along this opposite edge; call the new line CD.
>
> Line CD is parallel to AB.

1. Construct a line through point F that is parallel to line DE. Write the name of the new line you would draw.
answer: FG

2. Construct a line through point Z that is parallel to line XY. What is the name of the line you draw?
answer: ZK

3. Given line PQ and point M, construct a line through M parallel to PQ. Name the new line.
answer: ML

Objective 2

Constructing Perpendicular Lines

> EXPLANATION
> To draw a line that is perpendicular to a given line at a specific point, use a compass and a ruler. Follow these steps:
>
> 1. Place the compass point on the given point on the line and draw an arc that cuts the line at two points (call them A and B).
> 2. Keeping the same compass width, place the compass point on A and draw an arc above the line.
> 3. Without changing the width, place the compass point on B and draw another arc that intersects the first arc; call the intersection point P.
> 4. Use the ruler to draw a straight line through the original point and P. This new line forms a 90° angle with the original line, so it is perpendicular.

> WORKED_EXAMPLE
> Example: You have a line segment AB and need a perpendicular line through point B.
>
> 1. Put the compass point on B and draw an arc that meets AB at two points, say C and D.
> 2. Keep the compass width, place the point on C and draw an arc above the line.
> 3. With the same width, place the point on D and draw a second arc that meets the first one at point E.
> 4. Draw a straight line through B and E. The line BE is perpendicular to AB, forming a right angle at B.

1. You have a line segment XY drawn on your paper. Using only a ruler and a compass, construct a line that is perpendicular to XY at point X. List the steps you would follow.

answer: 1. Place the compass point on X and draw an arc that cuts XY at two points, say A and B. 2. With the same compass width, place the point on A and draw an arc above the line. 3. Without changing the width, place the point on B and draw another arc that meets the first arc at point P. 4. Draw a straight line through X and P. The line XP is perpendicular to XY.

2. A line LM is already drawn. You need to draw a line through point M that makes a right angle with LM. What construction steps do you use?

answer: 1. Place the compass point on M and draw an arc that cuts LM at two points, say Q and R. 2. Keep the compass width, place the point on Q and draw an arc above the line. 3. With the same width, place the point on R and draw another arc that intersects the first at point S. 4. Draw a straight line through M and S. The line MS is perpendicular to LM.

Objective 3

Bisecting a Line Segment

> EXPLANATION
> To bisect a line segment you need to find its exact midpoint using only a compass and a straightedge. Follow these steps:
> 1. Draw the given segment AB.
> 2. Place the compass point on A and draw an arc above and below the line with a radius more than half the length of AB.
> 3. Without changing the radius, place the compass point on B and draw another pair of arcs that intersect the first pair.
> 4. The two intersection points (one above, one below) are joined with a straight line.
> 5. This line cuts AB at its midpoint M, giving two equal parts AM and MB.

> WORKED_EXAMPLE
> Example: Segment AB is drawn 8 cm long.
> 1. Set the compass to a radius of about 5 cm (greater than 4 cm).
> 2. With the point on A, draw arcs above and below AB.
> 3. Keeping the same radius, draw arcs from B; they intersect the first arcs at points P (above) and Q (below).
> 4. Draw the line PQ.
> 5. PQ meets AB at point M, which is exactly 4 cm from each end, so AM = MB = 4 cm. Thus AB has been bisected.

1. You have a line segment CD drawn on paper. Explain how to bisect CD using a compass and straightedge.

answer: 1. Draw CD. 2. Place the compass on C and draw arcs above and below CD with a radius larger than half CD. 3. Without changing the radius, place the compass on D and draw arcs that intersect the first pair. 4. Join the two intersection points with a straight line. 5. The line meets CD at its midpoint, giving two equal parts.

2. Segment EF is 12 cm long. After bisecting it, what are the lengths of the two resulting parts?

answer: Each part is 6 cm long.

3. While bisecting a segment GH, you accidentally change the compass width when drawing the second set of arcs. Will the line you draw still pass through the true midpoint? Explain.

answer: No. The two sets of arcs must have the same radius; changing the width means the intersection points are not equidistant from the ends, so the line drawn through them will not cross the true midpoint.

How to Construct a Right Angle (90°) with a Compass and Ruler

EXPLANATION

A right angle measures exactly 90°. You can draw one using only a ruler and a compass. Follow these steps:

1. **Draw a base line** – Use the ruler to draw a straight line and mark a point O on it where the right angle will be placed.
2. **Set the compass** – Place the compass point on O and open it to any convenient radius (the length does not matter).
3. **Draw an arc** – With the compass still centred at O, draw an arc that cuts the base line at two points; call them A and B.
4. **Re-position the compass** – Without changing the radius, place the compass point on A and draw a small arc inside the first one. Repeat from point B to create a second small arc. The two new arcs intersect at point C.
5. **Draw the right-angle arms** – Use the ruler to draw straight lines from O through C and from O through the opposite side of the base line (the side opposite A and B). The two lines OC and the extension of the base line form a 90° angle at O.

Why it works: The arcs ensure that $OA = OB = OC$. Triangle OAC and OBC are congruent isosceles triangles, making $\angle AOC$ a right angle.

Now try it yourself!

WORKED_EXAMPLE

Worked Example

Suppose you have a line segment PQ and you need a right angle at point P.

1. Draw line PQ and mark P.
2. Set the compass to a comfortable width, place the point on P, and draw an arc cutting PQ at R and S.
3. Without changing the width, place the compass on R and draw a small arc inside the first one.
4. Move the compass to S and draw another small arc that meets the previous one at T.
5. Draw a line from P through T. The angle $RP T$ (or $SP T$) is exactly 90°.

Check: Using a protractor, you will see that the angle measures 90°, confirming the construction.

1. Using a ruler and compass, construct a 90° angle at point A on the line AB.
answer: A right angle with vertex A and arms AD and AE such that $\angle DAE = 90°$.

2. Given a line segment CD, construct a perpendicular line through point C.
answer: A line through C that forms a right angle with CD.

3. From point O, draw a 90° angle without any given line (freehand using compass).
answer: Two rays OA and OB such that $\angle AOB = 90°$.

Constructing a 60° Angle

EXPLANATION

You can construct a 60° angle with just a ruler and a compass by creating an equilateral triangle. An equilateral triangle has all three interior angles equal to 60°, so one of its sides will give you the required angle.

Steps to construct a 60° angle at point O on a given line l:

1. Draw a straight line l and mark a point O on it. This will be the vertex of the angle.
2. Place the compass point on O and draw an arc that cuts the line l at a point P.
3. Without changing the compass width, place the compass point on P and draw another arc that intersects the first arc at a point Q.
4. Draw a straight line from O through Q. The angle ∠POQ is 60° because triangle OPQ is equilateral (OP = OQ = PQ).
5. The two rays O-P and O-Q form the required 60° angle.

Remember to keep the compass width the same for steps 2 and 3; this ensures the triangle you create is equilateral.

WORKED_EXAMPLE

Worked example: Construct a 60° angle at point A on the line AB.

1. Draw line AB and mark point A on it.
2. Set the compass to any convenient radius and place the needle at A. Draw an arc that cuts AB at point C.
3. Keeping the same radius, place the needle at C and draw an arc that meets the first arc at point D.
4. Join A to D with a straight line.
5. The angle ∠CAD is 60°, because triangle ACD is equilateral (AC = AD = CD).

Check your construction: the three sides AC, AD, and CD should be equal in length, confirming the angle at A is 60°.

1. Construct a 60° angle at point O on a given line. List the steps you used in your construction.

answer: 1. Draw the line and mark point O. 2. With the compass set to any radius, draw an arc from O intersecting the line at P. 3. Without changing the radius, draw an arc from P intersecting the first arc at Q. 4. Draw a line from O through Q. 5. The angle between the original line and the new line (∠POQ) is 60°.

2. Given a line segment AB, construct a 60° angle at point B. Describe each step of your construction.

answer: 1. Extend AB if necessary and mark point B. 2. Set the compass to a convenient width and draw an arc from B intersecting the line at C. 3. Keep the same compass width, place the needle at C and draw an arc that meets the first arc at D. 4. Draw a line from B through D. 5. The angle ∠CBD is 60°, because triangle BCD is equilateral.

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	5.00	5.00	+0.00	+0.00
Taught-lesson alignment	4.00	4.00	+0.00	+0.00
Scope and conceptual correctness	4.00	4.00	+0.00	**+0.33**
Student-facing clarity and usefulness	4.00	5.00	**+1.00**	**+0.33**